Software Requirements Specification for Keystone: Capstone Project Management System

Team 13

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Revision History

Date	Version	Notes
Oct 13th 2026	1.0	Introduction and System Overview
Date 2	1.1	Notes

1 Introduction

This software requirements specification (SRS) document describes the system and product vision as well as the functional and non-functional requirements for Keystone. The SRS is intended to be a reference for the development team, stakeholders, and future maintainers of the system. This document is constructed using the IEEE 29148 and IEEE 15288 standards as a guide (**??**).

1.1 Purpose

The purpose of Keystone is to make capstone project proposals, discovery, team formation, and team project approval transparent, consistent, and easy to manage for the students, supervisors, and course instructors involved.

Capstone project and team selection currently happens across disconnected tools such as email, online forms, discussion forums, and shared files, leading to a lack of visibility, clarity, and detail in information available to students, supervisors, and instructors. As a result, project descriptions do not set clear expectations, students and supervisors cannot easily tell which projects are available or learn about each other, and instructors spend significant time manually coordinating the process. See Problem Statement for a more detailed description of the problem.

Keystone is intended to solve these problems by giving every participant a single, trustworthy source of information about capstone projects, teams, and their status, from the moment a project is proposed until it is approved.

1.2 Scope

Problem being solved. The fragmented and unreliable flow of information in capstone project and team selection and approval, as described in the Problem Statement section below.

Intended users. Capstone students, project supervisors (McMaster faculty and industry sponsors), course instructors, and teaching assistants (TAs), across the Software Engineering (SE), Computer Science (CS), and Mechatronics (TRON) capstone courses.

Major functionality. Keystone covers the lifecycle from project proposal to team project approval. Within that boundary the system:

- provides proposal templates for supervisors and for students, so that projects are described consistently;
- lets instructors review supervisors' project proposals before they are listed, and agree changes with the supervisors;
- provides a searchable catalog of projects showing which are open, taken, or able to take more teams;

- lets students and teams create structured profiles of their skills, experience, interests, and work style that other students and supervisors can view, indicate whether they are looking for a team, and find teammates interested in the same projects;
- suggests compatible teammates and suitable projects to students, based on their profiles and preferences
- lets student teams express interest in supervisor-proposed projects;
- supports communication between all users regarding project proposals, team formations, applications and and approvals, ensuring discussions are organized and accessible to relevant parties;
- supports comment threads on projects, student posts, and applications, so that questions and answers are kept in one place;
- lets supervisors and student teams each accept or decline working together, and records when both have agreed;
- notifies users of events that need their attention, such as new projects, status changes, and replies;
- provides supervisors with course process, timeline, and expectation information, and examples of past capstone projects, configurable by the course instructor;
- gives instructors visibility across courses into projects, teams, and their status, including which supervisors' projects were not selected, so that those supervisors can be informed;
- provides guidance and automation to help students meet all requirements when submitting their projects for approval, and when submitting changes to their project information;
- lets instructors approve, request changes to, or reject a team's project submission and later changes to its information;
- lets instructors configure which features are enabled and disabled for full flexibility

What Keystone does not do.

- It does not judge whether a project is feasible, sufficiently complex, or appropriate for a capstone course; this remains the responsibility of the course instructor.
- It does not make approval decisions. Final approval of teams and projects always rests with the course instructor. Courses may carry out approval inside the platform or in an external process.

- It does not resolve intellectual property (IP), confidentiality, or non-disclosure questions between students and supervisors.
- It does not find supervisors for student-proposed projects. Student-proposed projects are visible to all users of the platform, but Keystone provides no matching or request mechanism for supervisors.
- It does not schedule meetings or provide direct messaging between users. Meetings take place outside the platform, and discussion is limited to comment threads on posts and applications.
- It does not handle supervision after a project begins, such as feedback requests to supervisors, team performance concerns, or grading.

Expected inputs. Inputs are text-based project and profile information entered by users, and project status changes made by authorized users.

Expected benefits. Students can make more informed decisions on which projects to pursue, and find compatible teammates; supervisors know who is interested in their projects, learn about the students, and receive fewer repeated emails; and instructors spend less time on repetitive coordination and gain visibility into the whole process.

1.3 Stakeholders

- **Customer:** Course instructors for the SE, CS, and TRON capstone courses (Dr. Smith, Dr. Moradi, and Dr. Wassyng respectively). They decide which features their course uses and make the final project and team approval decisions.
- **End Users:**
  - Capstone students, who browse projects, create profiles, form teams, and apply to projects.
  - Project supervisors, who propose projects, manage their capacity, and review and confirm teams. Supervisors are either McMaster faculty or industry sponsors.
  - Course instructors and TAs, who manage users, projects, project status, and course configuration.
- **Development Team:** Ayush Patel, Waqar Ul-Hassan, Ahmed Rashrash, and Andrew Habib.

1.4 Definitions and Acronyms

Term	Definition

Capstone course	A final-year design course in which student teams spend roughly eight months (September to April) designing, implementing, and validating a system for a real group of stakeholders. In this document, the SE, CS, and TRON capstone courses.
SE, CS, TRON	Software Engineering, Computer Science, and Mechatronics Engineering, the three programs in the Computing and Software Department whose capstone courses Keystone supports.
Course instructor	The instructor responsible for a capstone course and for approving its teams and projects.
Supervisor	A McMaster faculty member or industry sponsor who proposes a project, or who agrees to advise a team on a project, and provides domain knowledge or acts as a client.
Industry sponsor	A supervisor from outside McMaster, who may have no McMaster account (MacID).
Project proposal	A description of a project submitted by a supervisor, or a student, for teams to consider.
Student-proposed project	A project proposed by a student team rather than by a supervisor.
Project status	The position of a project in its lifecycle, for example proposed, listed, taken, or approved.
Student profile	A student's skills, experience, interests, work style, and project preferences, visible to other students and to supervisors.
Team profile	A description of a team's members and shared goals, project interests, and work style.
Supervisor capacity	The number of teams a supervisor is willing to supervise on a project.
Team formation rules	Course rules on team composition: which programs may work together on which projects, and the minimum and maximum team size.
Proposal approval	The course instructor's review of a supervisor's project proposal before it is listed for students. The instructor may list it as submitted, work with the supervisor to make changes, or reject it.
Team project approval	The course instructor's decision to approve a team's selected project and membership
Approval mode	How a course handles approval: in-platform, external, or integrated.
Custom document	A deliverable, chosen by each team from a list, that suits the type of project (for example, a usability testing report).

TA	Teaching assistant assigned to a capstone team after its project is approved.
MacID	A McMaster University user account identifier.

Table 1: Definitions and Acronyms

1.5 Characteristics of Intended Reader

The intended reader of this SRS should have a background in software engineering, including knowledge of software development processes, requirements engineering, and system design principles. They do not have to be experts on these topics, but they should at least have the knowledge of a final year undergraduate student in software engineering. No knowledge of the capstone course process beyond the definitions in this section is required.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

Current situation. Capstone team and project selection is spread across disconnected tools, including email, online forms, MS Teams, Avenue to Learn and GitLab. Supervisors submit proposals through a form. The SE course instructor, who collects proposals for all three courses, reviews them, discusses any changes with the supervisor by email, and edits the list by hand before it is distributed to students as a static list. Students then form teams informally, mainly through discussion forums on Avenue to Learn / Teams and personal contacts, contact supervisors by email, and submit their chosen project and team for instructor approval through a course-specific process. No single place shows the state of the whole process.

Problems. The following problems were identified through interviews with stakeholders and general observation.

- **Project descriptions are inconsistent and do not set expectations.** Proposals vary widely in the level of detail, and very short descriptions put students off. Some of the things students want to know include a project's end goal and expected impact. Also, descriptions often do not state how much of the project already exists. Supervisors are unsure what a capstone project should look like and what a proposal should include, and students who pitch their own ideas do not know what a proposal needs. This makes projects difficult to compare and leads to mismatched expectations.

- **Student-proposed projects are hard to see.** Students' own project ideas appear only in discussion forums, where supervisors cannot see them, including supervisors who would like to supervise a student-proposed project. They are not tracked alongside supervisor-proposed projects.

- **There is no visibility of project status.** Students and supervisors cannot see which projects are taken or open, or whether a supervisor is willing to take more than one team, and after a rejection students do not know which projects remain. New projects and updates are hard to notice, and users do not want to have to remember to check for them.

- **Compatible teammates are hard to find.** Students without an existing network rely on forum posts or chance. Teammates' project preferences, skills, and work styles are often unknown until a team has formed, so teams can disagree about what to build.

- **Supervisors do not know the students' backgrounds.** Supervisors have little structured information about the students and teams who contact them, such as their skills, experience, and program. They can only gain this information through interviews or rely on outside sources such as LinkedIn profiles, which are inconsistent and not always available.

- **Communication during selection is inefficient.** Supervisors receive repeated emails from teams, including after a team has been chosen. What one team learns about a project in a meeting is not shared with other teams applying for the same project. Nobody can see which teams are in contact with which supervisors. There is also no record that a team and a supervisor agreed to work together. Students sometimes report that a supervisor agreed when the supervisor has no record of the team, so the instructor must confirm agreements separately.

- **Supervisors lack guidance before proposing and responding.** Before proposing a project, supervisors receive only a short summary of the course's expectations and process. The full course information arrives only after teams and projects are settled, and supervisors want it earlier, along with examples of past projects. Supervisors are also unsure how quickly to reply to students and whether to speak to every team that contacts them. This leads to misunderstandings about what a capstone project should deliver and to many questions to the instructor.

- **Instructors lack visibility across courses.** Instructors cannot see which projects students in other capstone courses selected, so they cannot tell which proposed projects were not selected. Supervisors whose projects are not selected are not informed, which may discourage them from proposing projects in future years.

- **Instructor review is repetitive and time consuming.** Instructors review each team's submission manually and often give the same comments to many teams on simple items, such as missing information, wasting the instructor's time. When reviewing

project proposals, changes to proposals that are agreed with supervisors by email must then be applied to the project list by hand.

- **Records drift after approval.** Team and project information changes after approval (for example, titles, supervisors, links, team composition, and TA assignments), but updating the official record takes extra steps, so it is often not updated and the instructor's records become out of date.

Impact. Students lose time early in a project of limited duration, teams end up mismatched with their projects or with each other, supervisors may become reluctant to propose projects in later years, and course instructors carry the administrative burden of the process.

Desired improvement. All participants share a single, accurate and current source of information about projects, supervisors' capacity, teams, and approval status; students can find suitable projects and teammates; supervisors can see students' skills and ideas and receive information about the course when they need it; and course instructors keep full control of approval decisions while spending less time on repetitive tasks.

2.2 Product Vision

Keystone will be the central place where capstone projects are proposed, discovered, and matched with student teams, and where instructors can see and manage the process. Each course decides which features it uses and how its approval process is carried out, so the platform can be adopted by different courses, departments, and universities.

The goals of Keystone are:

1. **Consistent project information.** Every project, whether proposed by a supervisor or by students, is described in a consistent format that makes its objectives, expected impact, required skills, current state, and expected outcomes clear.

2. **Visibility of availability and updates.** Users can tell which projects are open, taken, or able to take additional teams, and are made aware of new projects and changes without having to check for them.

3. **Easier team formation.** Students can describe their skills, experience, work style, and project preferences in a structured profile, find teammates with compatible profiles and preferences, and propose projects of their own for others to join.

4. **Informed matching.** Students receive suggestions of compatible teammates and suitable projects based on their profiles and preferences.

5. **Informed supervisors.** Supervisors can see student-proposed projects and the skills and experience of the students and teams interested in their projects.

6. **Efficient communication during selection.** Questions and answers about a project or application are kept in one place, supervisors can see which teams are interested in their projects, and fewer repeated emails are needed.

7. **Clear agreement between supervisors and teams.** A supervisor's and a team's agreement to work together is recorded in the platform and is visible to the instructor.

8. **Timely guidance for supervisors.** Supervisors have access to information about the capstone course, its timeline, and its expectations, and to examples of past projects, when they consider proposing a project, as configured by the course instructor.

9. **Oversight for instructors.** Instructors can see projects, teams, supervisors, and their status across courses, including projects that were not selected.

10. **Instructor-controlled approval with less repetitive work.** Final approval decisions remain with the course instructor. The platform supports the project proposal approval process, guiding supervisors to write more detailed project proposals and facilitating discussion between the instructor and supervisor. The platform supports the team project approval process carried out in the course, whether that takes place in the platform or elsewhere, helps students submit complete and correct information, and helps keep records current.

11. **Configurable and reusable.** Features can be turned on or off, and course-specific rules can be configured, so that the platform serves different courses without modification.

2.3 System Context

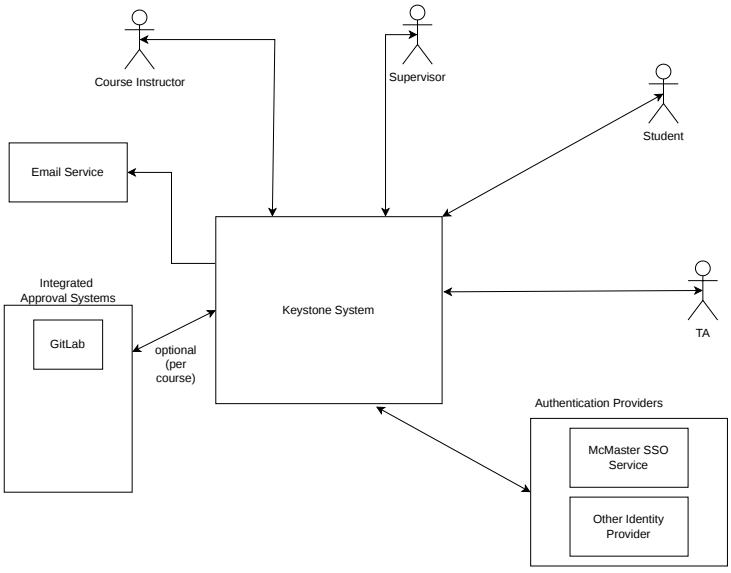


Figure 1: System Context

The entities that interact with Keystone are students, supervisors, course instructors, TAs, authentication providers, an email service, and the integrated approval systems of courses that do not carry out approval within Keystone.

- Student Responsibilities:
  - Provide an accurate profile including skills, experience, interests, work style, and project preferences.
  - Provide accurate team information and project proposals, including student-proposed projects.
  - Express interest in projects, take part in discussions, and respond to supervisors' decisions.
  - Submit complete project and team information for approval and keep it current.
- Supervisor Responsibilities:
  - Provide accurate, complete project proposals, including the current state of the project, expected outcomes, and the number of teams they are willing to supervise.
  - Review interested teams, take part in discussions, and accept or decline working with each team.
  - Keep their capacity and project information current.
- Course Instructor Responsibilities:
  - Configure the features, rules, and approval mode that apply to their course.
  - Make final decisions on teams and projects, including project proposal and team project approvals, requests for changes, and rejection.
  - Judge project feasibility and appropriateness for the course.
  - Provide the course process, timeline, and expectation information and the examples of past projects shown to supervisors.
- TA Responsibilities:
  - Update project and team status as directed by instructors, in courses where approval takes place outside Keystone.
  - Assist instructors in keeping records accurate.
- Authentication Provider Responsibilities:
  - Verify the identity of users who sign in, including users with and without McMaster accounts.
- Email Service Responsibilities:

  - Deliver the notifications generated by Keystone to users' email addresses.

- Integrated Approval System Responsibilities:
  - In courses that use one, carry out approval of teams and projects and make the outcome available to Keystone. Otherwise, instructors or TAs record the outcome in Keystone manually.

- Keystone Responsibilities:
  - Provide proposal templates, and check that required project, profile, and team information is present and well formed.
  - Show current project status, availability, and supervisor capacity.
  - Apply the team formation rules configured for each course.
  - Keep discussions attached to the project, post, or application they concern, and record whether a supervisor and a team have agreed to work together.
  - Restrict what each user can see and change according to their role.
  - Record who made each change to a project or team, and when.
  - Notify users of events that need their attention.
  - Provide instructors with an overview of projects, teams, and their status across courses.
  - Suggest compatible teammates and projects based on profile data.

Keystone is used to coordinate an academic process. Use is concentrated around the start of the academic year, when projects are proposed and teams are formed, with lighter use afterwards. It is not mission-critical or safety-critical. A failure delays the coordination of projects and teams but does not cause harm. The data handled includes personal information about students, so privacy is the main risk to be managed, rather than safety.

2.4 User Characteristics

- **Students** are final-year undergraduate students in the SE, CS and TRON programs. They are comfortable using web applications and should be able to figure out how to use Keystone without any training. They use Keystone mainly at the start of the course. They are motivated to find a suitable project quickly, but differ in whether they already have a team or a project idea.

- **Supervisors** are McMaster faculty or industry sponsors. They have a wide range of backgrounds. They may use Keystone only a few times a year, have limited time, and may be unfamiliar with the capstone course, its expectations, and its timeline. Industry sponsors may have no McMaster account. No technical training beyond using a web application can be assumed, and the effort needed to propose a project must stay low enough that it does not discourage them.

- **Course instructors** know the course's process and rules. They configure the platform for their course and make approval decisions. They use it throughout the term, and need to configure it without developer help.
- **TAs** are graduate students who know the course's process. They help instructors with day-to-day administration and are comfortable using web applications and technical tools.

2.5 System Constraints

- **Web-based.** Keystone shall be accessible through current versions of common web browsers on desktop, tablet, and mobile devices, and shall not require specialized hardware or installation. An internet connection is required.
- **Modular features.** Each major feature shall be able to be enabled or disabled independently for each course, so that a course can use only the parts that fit its process.
- **Independent of any one approval tool.** Keystone shall not depend on a specific external tool, such as the GitLab merge request process used by the Software Engineering course, to function. It shall support courses that carry out approval in the platform, courses that carry it out elsewhere, and, where an adapter exists, courses that synchronize with an external process.
- **Users outside the university.** Keystone shall support users who do not have a McMaster account (such as industry sponsors) alongside users who do.
- **Configurable course rules.** As Keystone should be extensible to different capstone courses, course-specific rules, such as which programs may work together on which projects and the minimum and maximum team size, shall be set by the course instructor as configuration and shall not be fixed in the design.

3 Stakeholder Requirements

SR1 Supervisors and instructors shall be able to propose a project using an structured template rather than a unstructured description. The template should include the expected scope/goals, the capstone team's deliverables, project impact, industry or field of study, confidentiality, and IP concerns. (S, SEI, ES)

SR2 The proposal template shall clearly distinguish required fields from optional ones, so that standardization does not discourage potential supervisors from submitting good projects and ideas. (SEI)

SR3 Supervisors and student teams shall have access to guidelines, an FAQ section, and examples of past capstone projects to help them generate ideas and judge appropriate size, scope, and content. (S, ES)

SR4 The course outline, calendar, objectives, and content-coverage expectations shall be available to potential supervisors before they propose a project. (ES)

SR5 Student teams shall have a documented process for proposing their own project and finding a supervisor for it - including a template and guidance on how to calibrate the scope. (S, IS)

SR6 Students can browse through all available projects in one place and see which projects are availible versus already assigned to a team. (S)

SR7 All stakeholders should be able to see the status of all project (e.g., open, taken, dropped, approved) at any time. (S, ES, SEI)

SR8 Stakeholders shall be notified when the status of a project they care about changes via email. (S, ES)

SR9 The instructors and TA's will be able to identify which projects have not been selected after project selection, in order to follow up with their supervisors. (SEI)

SR10 Approval decisions shall be made by the instructor and not automated. Routine checks and repeated feedback should be automated to reduce the instructors workload. (SEI)

SR11 Students and the instructor shall be able to exchange feedback iteratively during the approval process, preserving student exposure to review-based workflows. (SEI, S)

SR12 Students should be able to provide background information (team bio, interests, skills, experience, linkedln link) that supervisors can see before meeting a team. (ES, IS)

SR13 The instructor and TA's can get accurate, up-to-date, and validated team composition data (e.g., all students are in exactly one team). (SEI)

SR14 Supervisors working with teams across multiple capstone courses shall be able to see how the timelines and requirements differ between those courses. (IS)

SR15 Supervisors can see each team's members, team number, the point of contact, and assigned TA. (ES, IS)

SR16 Stakeholders shall be notified of tasks and deadlines through channels they already use (outlook), so that they do not have to remember to check for them. (ES)

SR17 Each stakeholder shall have a checklist of the actions required of them and their current completion state. (ES)

SR18 Stakeholders should be able to do capstone coordination (projects, schedule, teams, communication, and notes) in one place. (S, ES)

SR19 Administrative tasks including the generation of a list of approved projects, groups, selected projects, and supervisors will be automated. (SEI, ES)

4 User Personas

4.1 Persona 1

- Name:
- Description:
- Goals:
- Frustrations:
- Technical Ability:

4.2 Persona 2

- Name:
- Description:
- Goals:
- Frustrations:
- Technical Ability:

5 Use Cases

5.1 UC1: Example Use Case

Primary Actor

TODO

Preconditions

- TODO

Main Success Scenario

1. TODO
2. TODO
3. TODO

Alternative Flows

- TODO

Postconditions

- TODO

6 Functional Requirements

FR1: The system shall allow supervisors to create and submit project proposals using a structured template including a title, problem statement/background, objectives, skills/requirements, team deliverables, project impact, team capacity details, level of current completion, hardware/non-hardware, industry, information regarding confidentiality and IP concerns, and additional project terms.

Rationale: Standardizing proposals reduces inconsistencies in project descriptions helping sudents understand exactly what the project is about and allowing supervisors to clearly highlight their expections.

Fit Criterion: Supervisors can successfully fill out the defined fields with the appropriate information and submit a proposal for review.

FR2: The system shall distinguish required project proposal fields from optional ones preventing submissions if any required fields are not sufficiently filled.

Rationale: Keeping most fields optional would prevent supervisors from being discouraged to propose a project idea if they lack some details regarding the scope of their project. A projec that lacks a detailed scope can still be a good idea.

Fit Criterion: The system prevents stakeholders from submitting projects where mandatory fields are not sufficiently filled but submits a project that has mandatory fields filled even if some optional fields are missing.

FR3: The system shall allow student teams to create and publish a project proposal in a separate student-proposed project list using a structured template including the title, problem statement/background, and proposed solution/methodology.

Rationale: Students need a dcumented way to present their own project ideas if they choose not to select from the list of ideas from supervisors.

Fit Criterion: A student team can successfully create a project proposal, fill out the defined fields with the above information, and the proposal appears in a student-proposed project list.

FR4: The system shall allow instructors to review project proposals submitted by supervisors and approve, request, or provide feedback and request changes.

Rationale: Instructors need to verify that the proposed project is sufficient for a Senior-level capstone project and that the details provided are sufficiet for an understanding of the project.

Fit Criterion: A instructor can access a project proposal then accept, reject, or request changes successfully. Rejection or change requests are sent back to the supervisor for revision while accepted projects are included in the student project catalog.

FR5: The system shall provide a centralized catalog where all projects approved by the instructor can be found, filtered, and browsed based on details and availability.

Rationale: Students should be able to find projects and compare the varying opportunities with their team's skills and interests.

Fit Criterion: Users can view all the projects that were approved by the professor, search by title or keywords, and filter by the varying project fields. The filter and search options should also only return the relevant projects.

FR6: The system shall display the current availability status of each project in the catalog (open, taken, dropped, approved) and the maximum number of teams it can take.

Rationale: It is critical for students and supervisors to have clear visibility which projects are still available for selection. This gives students the ability to still request available projects if they have been rejected from others and supervisors to properly advertise their status.

Fit Criterion: Each project in the catalog clearly displays the availability status and the team capacity. Even if labels are not updated in real-time, the core application should respond to user actions appropriately based on status (i.e. taken projects cannot be requested).

FR7: The system shall allow students to create and update their profiles providing background information including bio, program, interests, project preferences, skills, experience, work style, and LinkedIn.

Rationale: Informative profiles allows students to discover other students with similar skills and interests when forming teams. It also allows supervisors to briefly evaluate interested students before interviewing them.

Fit Criterion: Students can create, save, update, and visit their profiles. These profiles should also be viewable by other students and supervisors.

FR8: The system shall allow students to create and join teams, invite specific students to join, recruit additional members, and assign roles.

Rationale: Students should be able to use the platform to create teams, join other teams, invite the people they know or fill in remaining spaces on their existing team by recruiting new individuals.

Fit Criterion: Students are able to create a team, send invitations to selected students to which those students can accept or deny, specify number of members needed and assign roles including a point-of-contact. Students can join an existing team.

FR9: The system shall validate team membership based on up-to-date course team formation rules, including minimum and maximum team sizes, permitted program combinations, and duplicate membership.

Rationale: The capstone course requires that certain constraints are met when forming teams and automating this reduces administrative efforts for TAs and Instructors.

Fit Criterion: The system prevents team formations that violate any of the rules mentioned above.

FR10: The system shall allow students who are seeking teammates to indicate their availability and discover other students or teams looking for members.

Rationale: Students who do not have a team or simply want to join a team instead of creating one will have a catalog of other teams and other students seeking others with similar interests.

Fit Criterion: A student can indicate whether they are looking for a team and they can discover other teams/students looking for additional members.

FR11: The system shall allow student teams to create and update a shared team profile including the member bios, collective skills, project interests, goals, and work style.

Rationale: Team profiles help supervisors evaluate the overal team's interests and whether the team's goals and skills align with the project in a consolidated way without having to go through each team member's profile.

Fit Criterion: Authorized team members can create and modify their team profile based on their team members' profiles. Other users can view these profiles.

FR12: The system shall recommend other available students/teams and capstone projects based on shared skills, interests, experience, and project preferences.

Rationale: Recommendations help student obtain the desired experience from the capstone course and work on projects that match their interests while pairing with like-minded students.

Fit Criterion: Students can receive teammate and project recommendations after their profile is complete based on highest similarities.

FR13: The system shall allow student teams to express interest in a supervisor-proposed project.

Rationale: A centralized application process allows supervisors to identify interested teams and communicate without relying on back-and-forth email chains.

Fit Criterion: A student can submit an application expressing their interest to the project that can be reviewed by the supervisor.

FR14: The system shall allow supervisors to accept or decline requests from student teams to work on their proposed projects, and record when both parties agree to the working arrangement.

Rationale: This reduces confusion by explicitely recording when both parties have understood each other's terms and have formally agreed to work with each other.

Fit Criterion: Both the supervisor and students can record a decision to confirm or deny the terms of the project. An agreement is marked after both parties accept.

FR15: The system shall allow any authorized users to create and reply to threads associated with their own project proposals, student team formations, and project applications. For TAs and Professors, that will be any projects in their scope.

Rationale: Users can exchange meaningful feedback, ask for clarification during the project selection stage, and communicate to understand project and course needs.

Fit Criterion: An authorized user can create or respond to a thread under a project proposal, project application, or a team formation they are associated with.

FR16: The system shall notify a user by email when there is a new reply in a thread they are part of, a decision on their application, agreement, or submission, or a status change on a project or team they are associated with.

Rationale: Notifications keep related stakeholders informed regarding project-related updates and decisions without repeatedly checking the platform.

Fit Criterion: Any changes to a project's status, decisions from either a student/supervisor, and thread updates, the system sends an email identifying the event to all relevant and authorized users. Students also get notifications for new project listing.

FR17: The system shall allow student teams to submit (or re-submit after modification given instructor feedback) their selected project and team information for instructor approval when in-platform approval is enabled.

Rationale: A centralized submission allows students to provide all capstone project related details for instructors to approve without relying on external tools.

Fit Criterion: Student teams can submit their project and team information through in-platform approval process that can be reviewed by instructors.

FR18: The system shall allow course instructors to approve, reject, or request modifications with feedback to submitted team-project applications when in-platform approval is enabled.

Rationale: Instructors must give the final says so it should not be an automated process.

Fit Criterion: An instructor can select an approval decision for the submitted team-project and provide feedback to improve project descriptions or identify missing items.

FR19: The system shall support recording capstone project and team approvals either manually by instructors or TAs associated with the project or automatically through an adapter.

Rationale: Different capstone courses have different approval systems that they would prefer to keep in place so the system must account for this.

Fit Criterion: For courses without integration adapter, either the TA or the instructor can manually update the status of the project as approved. If there is an adapter available, the system will automatically synchronize and update the approval status based on the information recieved from the external system.

FR20: The system shall validate submissions before sending them to instructors to ensure project information is complete, the project is available, the team does not already have another approved project, and the submission adheres to course-specific requirements.

Rationale: To reduce instructor workload, repeated feadback and routine checks based on course constriaints should be done automatically.

Fit Criterion: The system will prevent project submissions if any of the requirements above are not met, otherwise, the project can be submitted.

FR21: The system shall allow users to submit changes to previously approved team and project information that will still require the instructor to review before confirming.

Rationale: Details about projects may change later and require updates.

Fit Criterion: Student teams can propose changes to approved project or team record. The orginal approval remains in effect until the instructor accepts the new changes.

FR22: The system shall allow course instructors to configure the platform features including team formation, project approvals, and project proposals rules by disabling any undesired features for their course.

Rationale: Different capstone courses have different requirements and workflows, so instructors must be able to configure or customize the system to meet their course's needs.

Fit Criterion: An instructor can successfully enable or disable a supported features and define constraints that match their course's workflow.

FR23: The system shall allow course instructors to publish and maintain the course outline, course resources, FAQs, timelines, and past capstone examples for students and supervisors to view as reference.

Rationale: Early access to course expectations reduces confusion about project scope, deliverables, deadlines, and responsibilities for both the student and supervisor.

Fit Criterion: An instructor can add, edit, and publish course resources where students or supervisors and retrieve published resources.

FR24: The system shall provide users with a checklist of required actions and their completion status based on the course requirements.

Rationale: Stakeholders need a concise way to identify outstanding responsiblities and monitor their progress in the capstone approval procedure. This helps them keep on top of tasks and prevents them from missing critical deadlines.

Fit Criterion: Users can view a checklist containing their assigned actions and corresponding completion states. The system updates completion states for actions that are done and verified, such as submitting a project proposal or completing an approval submission.

FR25: The system shall send users administrative email notifications for upcoming deadlines and outstanding actions from a user's checklist.

Rationale: Email notifications will help stakeholders complete required tasks and fulfill their capstone responsibilities without regularly checking in the platform.

Fit Criterion: When configured, the system sends an email identifying incomplete or urgent action items and upcoming deadlines.

FR26: The system shall provide authorized Instructors and TAs with access and detailed information for projects, teams, supervisors, and their statuses across their assigned capstone course(s) or designated capstone teams.

Rationale: Instructors need visbility to identify unselected projects, coordinate with supervisors, and maintain accurate admin details.

Fit Criterion: An authorized instructor or TA can view and filter project and team statuses, monitor approval progress, and access relevant discussions.

FR27: The system shall provide an automatically generated exportable report to Instructors and TAs providing a list of approved projects, team compositions, project selections, and supervisors.

Rationale: Automated reports reduce manual administrative work for Instructors and TAs if they want to obtain accurate course-wide records through compiled documentation.

Fit Criterion: An authorized instructor or TA can generate and download reports containing selected admin records reflecting approved projects, team compositions, project selections, and supervisor information.

FR28: The system shall allow users to authenticate via McMaster Single Sign-On (SSO) for McMaster students and staff or another trusted authentication system for external supervisors before accessing any other services.

Rationale: This allows the platform to uphold its security when handling sensitive information and that the account is accessed by the intended individual.

Fit Criterion: A user can sign in using an available identify provider and access the functions available to them.

FR29: The system shall restrict access to system functionality and information according to the user's role and course permissions.

Rationale: Different stakeholders (Students, TAs, Instructors, and Supervisors) have different roles in the platform so access restrictions are critical to prevent unauthorized

access when it comes to viewing and modifying content. An example is that teams should only have access to modify only their team's approved project details. An Instructor can modify any project's details.

Fit Criterion: Users are only able to access information and perform actions permitted by their assigned roles and scope.

FR30: The system shall allow authorized administrators to manage users' course association and assigned roles.

Rationale: Different capstone courses involve different participants, and access permissions must reflect each users' responsibilities.

Fit Criterion: An authorized administrator can view and update a user's course association and permitted course role.

7 Data Requirements

7.1 Major Data Entities

- User
- Song
- Session
- Report

7.2 Data Dictionary

Attribute	Type	Description
user_id	UUID	Unique identifier
email	String	User email address

Table 2: Data Dictionary

7.3 Privacy and Data Retention

TODO

8 Non-Functional Requirements

NFR1: **Accuracy**

NFR2: **Usability**

NFR3: **Maintainability**

NFR4: **Portability**

- Other NFRs that might be discussed include verifiability, understandability, reusability, performance, reliability and availability, security, and accessibility. You may have to add other NFRs.

9 AI and Machine Learning Requirements

9.1 Model Inputs

TODO

9.2 Model Outputs

TODO

9.3 Evaluation Metrics

TODO

9.4 Known Limitations

TODO

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR??	UC1	FR1

Table 3: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req

FR1	NFR1

Table 4: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

12 Likely Changes

LC1:

13 Unlikely Changes

LC2:

14 Appendices

14.1 Appendix A: Additional Materials

14.1.1 Stakeholder Interview Notes

The notes below were taken during stakeholder interviews. Each note is numbered with its stakeholder tag (S, ES, IS, I) so that stakeholder requirements can reference the specific note(s) they came from.

Student (S) Student who chose a project from the list of given projects.

S1 Formed a team with classmates they already knew over the summer, and had project ideas before the term started. In hindsight, picking the project before the teammates might have been better, since teammates with shared interests working on their preferred project could lead to a better experience.

S2 Each member chose their top 5 projects after reading all of them, then the team settled on a top 3 in case some were unavailable.

S3 Project descriptions were often vague. Some were ignored and only the title was read.

S4 Information on hardware, software stack, and industry/field of study would be helpful.

S5 One supervisor held a meeting with all interested groups right before the project selection deadline, which added uncertainty.

S6 Another group was competing for the same project, and losing it led to frustration.

S7 Considerable uncertainty while landing a project, with worry about not having one available.

S8 A template for proposing their own project would help.

S9 Guidelines or an FAQ on proposing their own project would reduce uncertainty about size, scope, and content requirements.

S10 Wants a way to know when a project is taken, open, dropped, etc.

S11 A notification mechanism for project status would help; a newsletter was suggested.

S12 Supervisor communication (mechanism, process, content) is unclear: is an agenda needed, will the supervisor mostly speak, who leads each meeting? Guidelines may be enough.

S13 Unsure whether supervisors will have time to meet regularly (e.g., weekly).

S14 Meetings are useful for answering a backlog of questions and filling knowledge gaps with an expert instead of researching alone.

S15 Prefers email for simple questions.

S16 "Suggested meetings" aligned with deliverables would set expectations and help groups know whether they are on track.

S17 The GitLab approval process was intuitive due to prior GitHub/GitLab experience.

S18 JSON files were a hassle but not problematic.

S19 Finding where things are is time consuming and requires many open tabs.

S20 Google Calendar was helpful but would benefit from more information.

S21 Having everything in one place (like Avenue) would have been more helpful.

External Supervisor (ES) Supervisor from outside the university.

ES1 The proposal process was smooth, and working with one of the professors and meeting the team was smooth as well.

ES2 Knowing project statuses is a challenge.

ES3 A centralized platform for meeting notes would help. Some teams get more information from meetings than others, and notes should be shared with everyone. Summaries of meetings and project status, and a template for them, would be great.

ES4 A schedule would clear up expectations. Some teams come in early and some late; timelines and deadlines are unclear.

ES5 A template would be better than an open-ended description box for proposals. It is hard to know what is expected from the capstone team, so a best-effort approach is taken.

ES6 Guidelines on course content, content coverage, and capstone objectives would let supervisors tailor the experience. The course calendar and outline are wanted earlier, ideally before proposing.

ES7 Examples (e.g., past capstones) would help with idea generation and expectation setting. Ideas were thrown around and refined internally, but guidance would have been better.

ES8 Extra back-and-forth with the professor (capstone processes, expectations, IP questions) could be avoided by providing more information up front; a template or FAQ would help.

ES9 Being notified to do tasks on a platform is fine; having to remember to open it is the problem. Email or phone notifications should be considered. A notification and checklist is preferred over searching in the platform.

ES10 Too many platforms to log in to is tedious. A new application is fine.

ES11 Suggested meeting times and a schedule would help, though some supervisors have their own schedule that works better.

ES12 No information about students before team meetings; a leap of faith was required, relying on gut feeling and the right questions. Did not know they had the final say.

ES13 Wants a way to learn about students (team bio, interests, skills, experience), even if not all shown up front; looking them up (e.g., LinkedIn) was mentioned.

ES14 Wants teams matched to projects based on preference criteria and skill sets.

ES15 Hard to find where to communicate back to students. Groups usually had one representative email and CC their members, which made emails hard to search. With many names and no group numbers, they defaulted to the first person who emailed. Inviting everyone was too taxing; wants some way to streamline.

ES16 Overall, there is a lot of administration on the supervisor side.

Internal Supervisor (IS) Professor at the university who supervises teams, submits proposals, and advises the instructors.

IS1 Mostly a supervisor. Submits proposals and then waits to see what else is on the list in case something interests him. Interviews and supervises groups, attends lectures when possible, helps with presentations and Expo posters, and is often on the outside of the course.

IS2 Unsure how student-proposed projects will be handled.

IS3 The three capstone courses are not well coordinated. The timing and method of the approval process (whether a project can be a capstone project) is not really known.

IS4 Current descriptions are 100–200 words. Standardizing them is hard and he is unsure what the standard should be. Some supervisors may want something specific, and one group delivered something related to, but very different from, the proposal.

IS5 Supervisors do not know the course schedule (when the SRS is due, when implementation starts, validation testing, user testing). There is confusion about what happens and when things are due.

IS6 Supervisors with teams across multiple courses (SE/CS/Tron) may not know that each course has different timelines and requirements.

IS7 No standardized procedure, e.g., how a group with an idea finds a supervisor. Wants the timeline and procedure centralized and standardized.

IS8 Wants a low-effort mechanism for supervisors to submit a report or feedback at the end. One professor sends feedback emails, but others do not seem to remember them and send reports voluntarily.

IS9 Communication and regular updates (check-ins) between supervisors and teams are essential and need to be accounted for. Discussion is needed on weekly meetings and who questions go to (TA or professor).

IS10 Wants to know who each of his groups' TAs are.

IS11 Student skills would help guide discussions with supervisors.

IS12 Would like the capstone process all in one place, but is unsure whether that would work with other professors, or whether the platform should cover only the proposal stage.

IS13 Accounts are mentioned as a need (details not captured in notes).

Software Engineering Instructor (SEI) Professor who runs the course.

SEI1 Current process: before the course, emails contacts to find projects. Interested supervisors submit an MS Form (title, organization, contact, description, confidentiality, IP concerns). A Google Form funnels into a spreadsheet that a script turns into LaTeX. Students form teams, choose projects, and organize meetings. A merge request in GitLab counts as approval.

SEI2 Supervisor details often change and are updated. Supervisors are contacted to confirm they know a team will work with them, which often surfaces Expo and other capstone details.

SEI3 Teams are classified by project hardware (SE vs. Mech).

SEI4 Team details are recorded in a JSON file, processed and verified by a Python script (e.g., every person is in exactly one team). The team composition CSV should be an output that is up to date, correct, and valid.

SEI5 Would rather spend time on decisions such as project viability and approval.

SEI6 Often has to reach out to potential supervisors to have enough projects each year.

SEI7 Wants to know which projects have not been selected, e.g., to send a thank you email.

SEI8 Coordination, communication, and details such as approved projects and groups should be automated as much as possible.

SEI9 Approval criterion: enough substance. Game projects often need extra work to add substance (e.g., more analysis such as proving fairness, or more components).

SEI10 Standards could be considered, but enforcing them can scare away good projects; optional questions may help.

SEI11 No AI agents making approvals. AI-generated summaries could be considered, but their usefulness is unclear.

SEI12 Pull requests are still favored because they expose students to CI and show its usefulness, and allow back-and-forth between professor and students.

SEI13 The same feedback is sent to multiple groups. An AI agent could respond to PRs with already-resolved issues (e.g., professor not added as collaborator), leaving more important work to the instructor.

SEI14 Possibly usable universally, which may require modularity and extensibility.

SEI15 Algorithmic matching of teams to projects could be interesting, with a way to turn it off. Overhead must be worth it; this could be offset by moving other things (e.g., meetings) from Teams or Zoom onto this platform.

14.2 Appendix B: Development Plan

14.3 Appendix C: Values of Auxiliary Constants

TODO

14.4 Appendix D: Generative AI Disclosure

14.5 Appendix E: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?